

Bishop's University

Department of Computer Science

Programming Languages fro Data Analysis - CS405

Assignment 1 - Covid19 Data Analysis

First:

1. Create a folder named **Assigment_1** in your GDrive.
 2. Inside The folder **Assigment_1**, create a Python Notebook named **A1_Group_XX.ipynb**, where XX is the number of your group.
 3. Load the file **time_series_covid19_confirmed_global.csv** from "CSSEGISandData" Github repository.
 4. Upload it into your GDrive **Assigment_1** folder.
 5. Answer the following questions.
 6. Upload your **A1_Group_XX.ipynb** on Moodle.
 7. Dealine: 30 Spetmeber (firm).
- Below are the results. You must provide the code in the corresponding cells that generate these results.

1. Mount your Google drive

2. Move to the **Assigment_1** (Change directory)

- This allow you to access your csv file

3. Load global data from csv file

In []: `# complete this cell`

Out[]:

	Province/State	Country/Region	Lat	Long	1/22/20	1/23/20	1/24/20	1/25/20	1/26/
0	NaN	Afghanistan	33.0000	65.0000	0	0	0	0	
1	NaN	Albania	41.1533	20.1683	0	0	0	0	
2	NaN	Algeria	28.0339	1.6596	0	0	0	0	

3 rows × 76 columns

4. Count and display the total number of NaN values if any

In []: `# complete this cell`

Out[]:

178

5. Replace the NaN values with the text 'not available'

- Show some rows with updated text

In []: `# complete this cell`

Out[]:

	Province/State	Country/Region	Lat	Long	1/22/20	1/23/20	1/24/20	1/25/20	1/26/20
0	not available	Afghanistan	33.0000	65.0000	0	0	0	0	0
1	not available	Albania	41.1533	20.1683	0	0	0	0	0
2	not available	Algeria	28.0339	1.6596	0	0	0	0	0
3	not available	Andorra	42.5063	1.5218	0	0	0	0	0
4	not available	Angola	-11.2027	17.8739	0	0	0	0	0

5 rows × 76 columns

6. Load only China's data

In []: `# complete this cell`

Out[]:

	Province/State	Country/Region	Lat	Long	1/22/20	1/23/20	1/24/20	1/25/20	1/26/20
49	Anhui	China	31.8257	117.2264	1	9	15	39	0
50	Beijing	China	40.1824	116.4142	14	22	36	41	0
51	Chongqing	China	30.0572	107.8740	6	9	27	57	0

3 rows × 76 columns

7. Update the index

- If the indexes dont start from 0, then reset it to start from 0.
- You may need a new dataframe.

In []: `# complete this cell`

Out[]:

	Province/State	Country/Region	Lat	Long	1/22/20	1/23/20	1/24/20	1/25/20	1/26/20
0	Anhui	China	31.8257	117.2264	1	9	15	39	0
1	Beijing	China	40.1824	116.4142	14	22	36	41	0
2	Chongqing	China	30.0572	107.8740	6	9	27	57	0
3	Fujian	China	26.0789	117.9874	1	5	10	18	0
4	Gansu	China	37.8099	101.0583	0	2	2	4	0

5 rows × 76 columns

8. Discard Country/region, Lat and Long columns

- Creat a new dataframe

In []: `# complete this cell`

Out[]:

	Province/State	1/22/20	1/23/20	1/24/20	1/25/20	1/26/20	1/27/20	1/28/20	1/29/20	1/30/20
0	Anhui	1	9	15	39	60	70	106	152	170
1	Beijing	14	22	36	41	68	80	91	111	125
2	Chongqing	6	9	27	57	75	110	132	147	160
3	Fujian	1	5	10	18	35	59	80	84	90
4	Gansu	0	2	2	4	7	14	19	24	28

5 rows × 11 columns

9. Transpose the table. Provinces as columns and Cases as rows

In []: `# complete this cell`

Out[]:

	Anhui	Beijing	Chongqing	Fujian	Gansu	Guangdong	Guangxi	Guizhou	Hainan	Hebei	Henan	Hubei	Inner Mongolia	Jilin	Shaanxi	Shandong	Shanghai	Sichuan	Tianjin	Yunnan	Zhejiang
0	1	14	6	1	0	26	2	1	4	1	1	1	1	1	1	1	1	1	1	1	1
1	9	22	9	5	2	32	5	3	5	1	1	1	1	1	1	1	1	1	1	1	1
2	15	36	27	10	2	53	23	3	8	2	1	1	1	1	1	1	1	1	1	1	1
3	39	41	57	18	4	78	23	4	19	8	1	1	1	1	1	1	1	1	1	1	1
4	60	68	75	35	7	111	36	5	22	13	1	1	1	1	1	1	1	1	1	1	1

5 rows × 22 columns

10. Data analysis

1. Display the number of cases by province during the 70 days considered

In []: `# complete this cell`

```
Anhui          57547
Beijing        26713
Chongqing      34445
Fujian         18269
Gansu          6646
Guangdong      81716
Guangxi        14892
Guizhou        8305
Hainan         9924
Hebei          17977
Heilongjiang   26835
Henan          74565
Hong Kong      11193
Hubei          3572897
Hunan          60437
Inner Mongolia 4548
Jiangsu        36696
Jiangxi        54546
Jilin          5385
Liaoning       7610
Macau          971
Ningxia        4232
Qinghai        1127
Shaanxi        14676
Shandong       41325
Shanghai       22190
Shanxi         7954
Sichuan        31210
Tianjin        8088
Tibet          64
Xinjiang       4283
Yunnan         10611
Zhejiang       73711
dtype: object
```

1. Describe the data

In []: `# complete this cell`

Out[]:

	Anhui	Beijing	Chongqing	Fujian	Gansu	Guangdong	Guangxi	Guizhou	Hainan	Hek
count	72	72	72	72	72	72	72	72	72	
unique	33	62	37	43	32	61	34	25	26	
top	990	399	576	296	91	1352	252	146	168	3
freq	35	3	27	22	17	4	21	45	44	

4 rows × 33 columns

